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| **Fiscal Year:** | FY 2021 | **Task Last Updated:** | FY 05/27/2021 |
| **PI Name:** | Pluth, Janice Ph.D. | | |
| **Project Title:** | Microgravity Effects on Mammary 3D Tissue Architecture and Role of Stem Cells and Genetics | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Animal Biology: Vertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Developmental Biology | | |
| **Space Biology Special Category:** | (1) Cell Culture<br>(2) Translational (Countermeasure) Potential | | |
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| **Zip Code:** | 89154-9900 | **Congressional District:** | 1 |
| **Comments:** | NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons |
| **Start Date:** | 07/26/2018 | **End Date:** | 07/25/2022 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA KSC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 7/25/2022 per NSSC information (Ed., 9/19/21) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Parvin, Bahram Ph.D. ( University of Nevada, Reno ) | | |
| **Grant/Contract No.:** | 80NSSC18K1464 | | |
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| **Performance Goal Text:** | | | |

**Task Description:**

During space travel, astronauts will experience a unique environment that includes continuous exposure to microgravity and stressful living conditions. Physiological adaptation to this is a challenge and the effect of microgravity on organ development, architecture, and function is not well understood. How microgravity may impact the growth and development of an organ is an important issue, especially as space flight becomes more commonplace. These fundamental questions can be addressed by studying simulated microgravity exposed cells in vitro. 3D tissue culture models mimicking mammary gland structures have been optimized and can be used to define if exposed cells can correctly assemble into these structures to address some of NASA's key questions regarding microgravity effects on cells and organs. Additionally, as proper activation of stem cells is key to organ formation it is important to define how microgravity may affect these specialized cells. In the proposed work we will expose a mammary epithelial cell line (Comma 1D) known to contain a high proportion of stem cells and quantify the changes in mammary stem cell populations before and after microgravity exposure. Further, we will identify how exposure to simulated microgravity affects the cells' ability to form normal 3D structures important in mammary organ development following this exposure. In addition, we will test role of genetics by isolating mouse mammary epithelial cells from mice showing increased and reduced sensitivities to cancer to define if simulated microgravity shows differential effects dependent on underlying genetics. Next, we will identify miRNA changes in stem cells isolated pre and post simulated microgravity exposure to define the underlying molecular reasons for defects in correct 3D structure formation. A final aim will be to define whether microgravity in combination with certain genetic backgrounds causes cellular changes indicative of an increased cancer risk. We hypothesize that cells exposed to simulated microgravity and assayed in 3D will allow us to define how this exposure would impact the growth and development of the mammary gland in vivo. We furthermore hypothesize that the magnitude of this response depends upon genetic susceptibility and thus will utilize cells from different mouse strains to investigate this (cancer prone and resistant). Lastly, we hypothesize that the simulated microgravity exposure in an at risk genetic background may cause changes in cells that increase the propensity to develop cancer. These studies will help identify the cellular and molecular changes and underlying genetic mechanisms of simulated microgravity and how it impacts overall tissue function.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

**Task Progress:**

The studies for this project were designed to address whether simulated microgravity affects mammary gland development using complex 3D tissue models. The majority of prior microgravity cell studies have focused on terminally differentiated cells or used immortalized cell lines, which do not provide a complete understanding of the in vivo situation. In these studies, we used a mouse cell line from a pregnant mouse, COMMA-D1, that contains higher numbers of progenitor and stem cells to better discern how stem cells may be influenced by microgravity. In addition, we used freshly isolated organoids from two very diverse strains of mice to provide a more realistic picture of microgravity effects on normal cells and to discern the role of genetics on microgravity effects.
In this third year of the project, we have continued to obtain additional data, and in some cases further optimized our assays to allow completion of the two stated AIMS:

AIM 1: To define the effects of simulated microgravity on stem cell populations in normal mouse mammary epithelial COMMA-D1 cells in 2D and their ability to form proper 3D tissue structures.

AIM 2: To investigate the role of genetics in modulating simulated microgravity effects on mammary stem cells and their ability to differentiate and form proper 3D structures, and to exhibit changes suggesting an increased carcinogenic potential.

Unfortunately, the pandemic has had a serious impact on our ability to complete the goals as scheduled and thus we would like to request a no cost extension to continue with this work. We are just beginning to get back to normal operations at the University, but we have continued with the work as best we could during the pandemic. Below is a summary of the work we have accomplished since the last annual report in 2020.

Aim 1 Results to date.

Exposure to Simulated Microgravity Changes Cell Cycle Profiles and Induces Apoptosis. Mouse mammary COMMA-D1 cells were exposed in 2D for six days. Following this exposure we noted as has been previously published that some cells continued to grow adherently (ug-adherent population) and some now grew in suspension (ug-non-adherent population). In comparing cell cycle profiles for these cells we noted that exposure to simulated microgravity changed the percentage of cycling and apoptotic cells as compared to control cells. In addition, cell diameter was also changed under simulated microgravity as compared to control. These results indicate that simulated microgravity can affect proliferation, cell size, and apoptosis.

Changes in Mammary Epithelial Cellular Markers Following Simulated Microgravity Exposure. We exposed COMMA-D1 cells in 2D to simulated microgravity and then analyzed control, ug adherent, and ug non-adherent populations for stem cell markers. We used two different stem cell staining protocols that have been published to identify MASC (mammary stem cells) and MRU (mammary repopulating units) populations. With simulated microgravity treatment and staining for CD24 vs CD49f populations it appeared that treatment changed the cellular markers using both staining protocols. We are currently staining the populations for other markers to determine how simulated microgravity may specifically modify stem cell populations.

As stem cells have the ability to grow in a non-adherent manner it is likely that a greater proportion of the non-adherently growing cells are stem or progenitor cells. We also tested each population’s ability to form 3D acini structures and discuss how simulated microgravity affects this endpoint below.

Optimization of Imaging Protocols for studying changes in 3D tissue growth. During the last year (May 2020-2021), our collaborators at University of Nevada, Reno (UNR) performed extensive refinements and validation of a method that they developed for delineation and profiling organoids in the previous year. They designed a new cost function, developed a better optimization, and demonstrated improved performance with the prior art. In addition, they defined a new set of indices that capture colony organization in terms of elongation, convexity, and surrogate indices for lumen formation. They routinely received image stacks from the Pluth Laboratory, processed them, and returned tabulated computed indices back to the University of Nevada, Las Vegas (UNLV). The new algorithm developed was included in a recent publication.

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| | Unique differences are observed in 3D tissue growth when mammary cells are grown under simulated microgravity. COMMA-D1 cells were exposed to simulated microgravity and grown in 3D tissue cultures to form acini structures, a normal structure of a typical mammary organ. Following microgravity exposure significant differences in acini formation were noted as compared to control. Six independent experiments have been completed to date and reveal differences between simulated microgravity treated cells as well as between adherent and non-adherent microgravity treated cells. We are currently performing additional experiments to obtain sufficient numbers for each treatment group.<br><br>Aim 2 Results to date.<br><br>Optimizing Organoid Preparation and Single Cells Recovery from Mouse Mammary Glands for 3D Mammary Acini Growth. Protocols have been optimized for obtaining the highest recovery of organoids and dissociated single cells for the mammary acini assay from fresh mouse mammary glands. Following optimization of single cells from the glands, further optimization of the acini assay using cells from each strain were begun. There was some difficulty in getting optimal acini growth from non-immortalized cells, but eventually with proper medium formulation we have preliminary results that reveal acini develop using primary mammary cells isolated from a mouse gland, although a lower number of acini develop based on equal seeding density as compared to immortalized cells. We are now ready to move ahead with additional studies using glands from both the C57BL6 and BALB/c mouse strains to define how genetic differences may impact the effects of simulated microgravity on 3D acini growth.<br><br>Branching Morphogenesis Assay. This is another 3D tissue growth type assay that mimics the growth of ductal branches within the mammary gland. Unfortunately, this assay has been a bit more difficult to optimize for the two mouse strains we are using for this study. We have been able to optimize conditions for one mouse strain and then worked to optimize conditions for the second strain. It required a slightly different formula and we are now attempting to test this on a larger scale with both strains in combination with exposure to simulated microgravity. Thus far we have some results from one strain of mouse, the C57BL/6 strain. From this preliminary work we note that the number of organoids seeded that branched is roughly the same between control and simulated microgravity treated cultures. However, there are slight differences in length, width, and branch length between the groups. Further studies will be performed to obtain a sufficient “n” for statistical comparison to determine if this trend proves to be significant. |
| **Bibliography Type:** | Description: (Last Updated: 11/01/2023) |
| **Articles in Peer-reviewed Journals** | Winkelmaier G, Parvin B. "An enhanced loss function simplifies the deep learning model for characterizing the 3D organoid models." Bioinformatics. 2021 Sep 15;37(18):3084-5. https://doi.org/10.1093/bioinformatics/btab120 ; PMID: 33620423; PMCID: PMC847966 , Sep-2021 |